G.2. Qwen3-14B Results

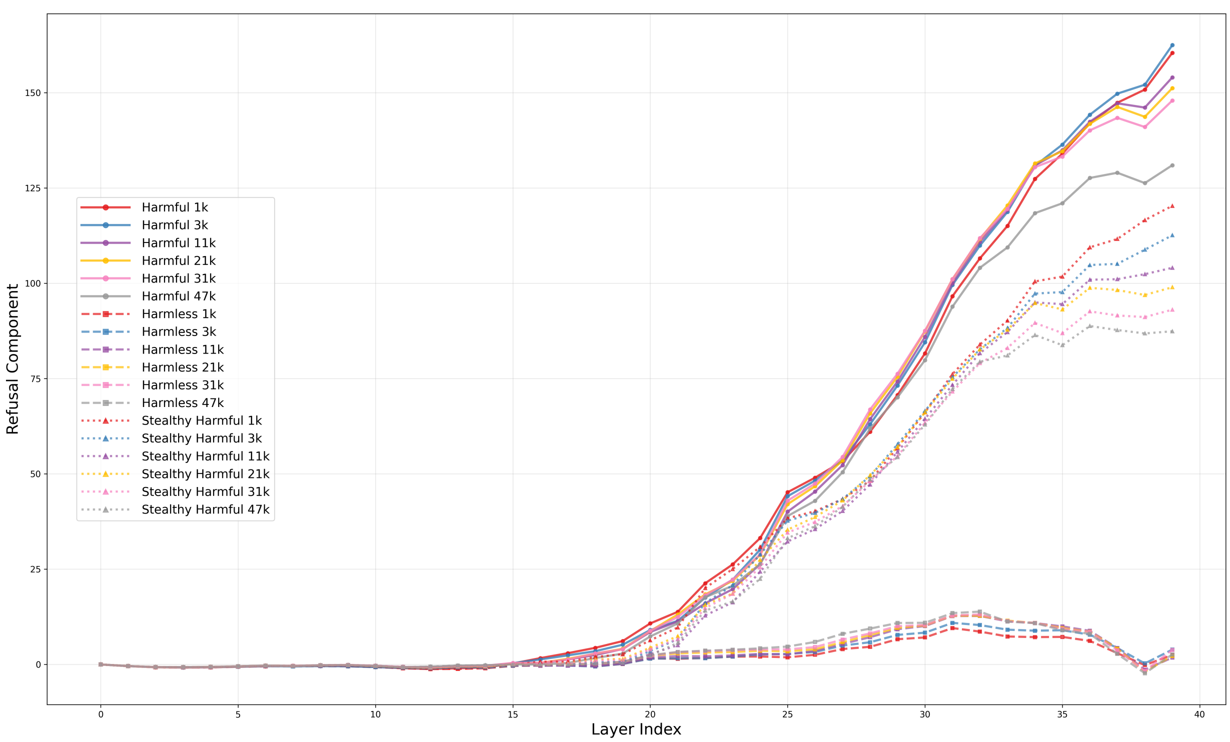

Figure 15. **Refusal component comparison across layers (Qwen3-14B) for different CoT lengths.** Includes harmless, harmful, and stealthy harmful instructions. CoT lengths: 1k, 3k, 11k, 21k, 31k, 47k tokens.

G.3. GPT-OSS-20B Results

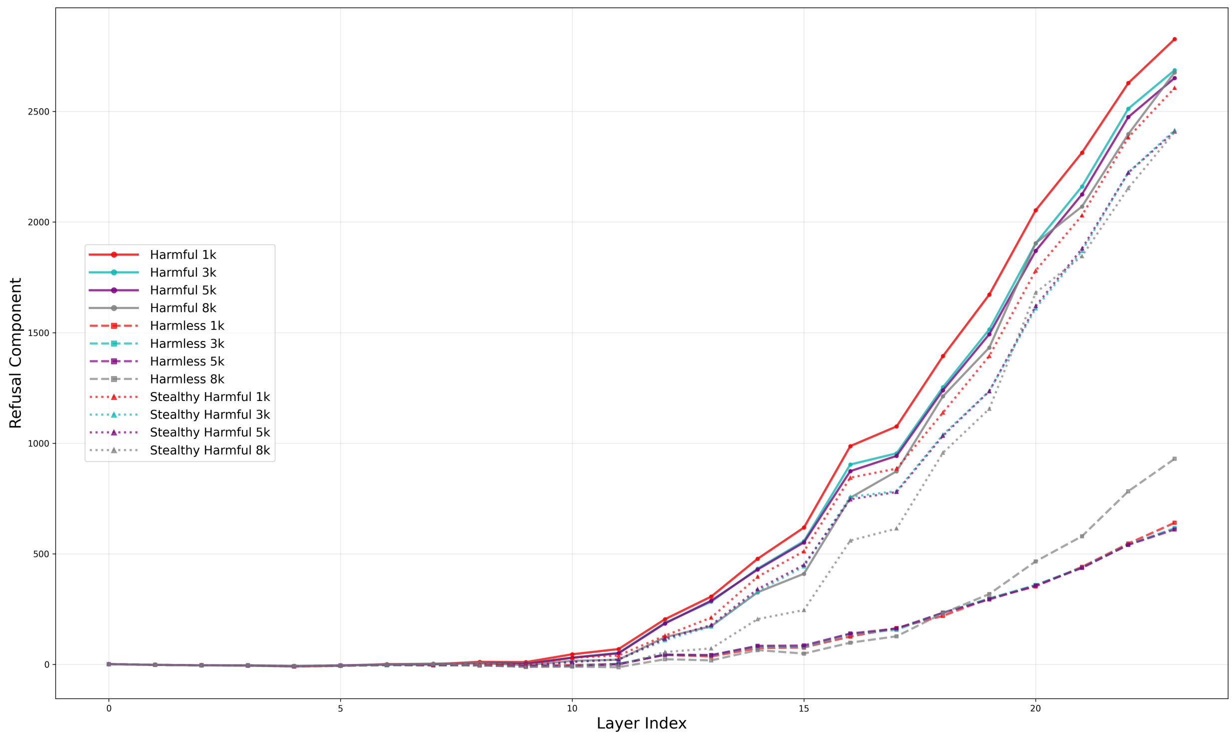

Figure 16. **Refusal component comparison across layers (GPT-OSS-20B).** Four template lengths: 1k, 3k, 5k, 8k tokens. Results show the same trend: longer CoT reduces refusal activation. Note that due to GPU memory constraints, the template length range is much narrower than in the Qwen3 experiments, yielding a less pronounced but consistent pattern.

H. Attention Ratio Analysis

$$\text{AttnRatio} = \frac{\sum_{t \in H} \alpha_t}{\sum_{t \in P} \alpha_t}, \quad \alpha_t = \text{attention weight on token } t \tag{3}$$

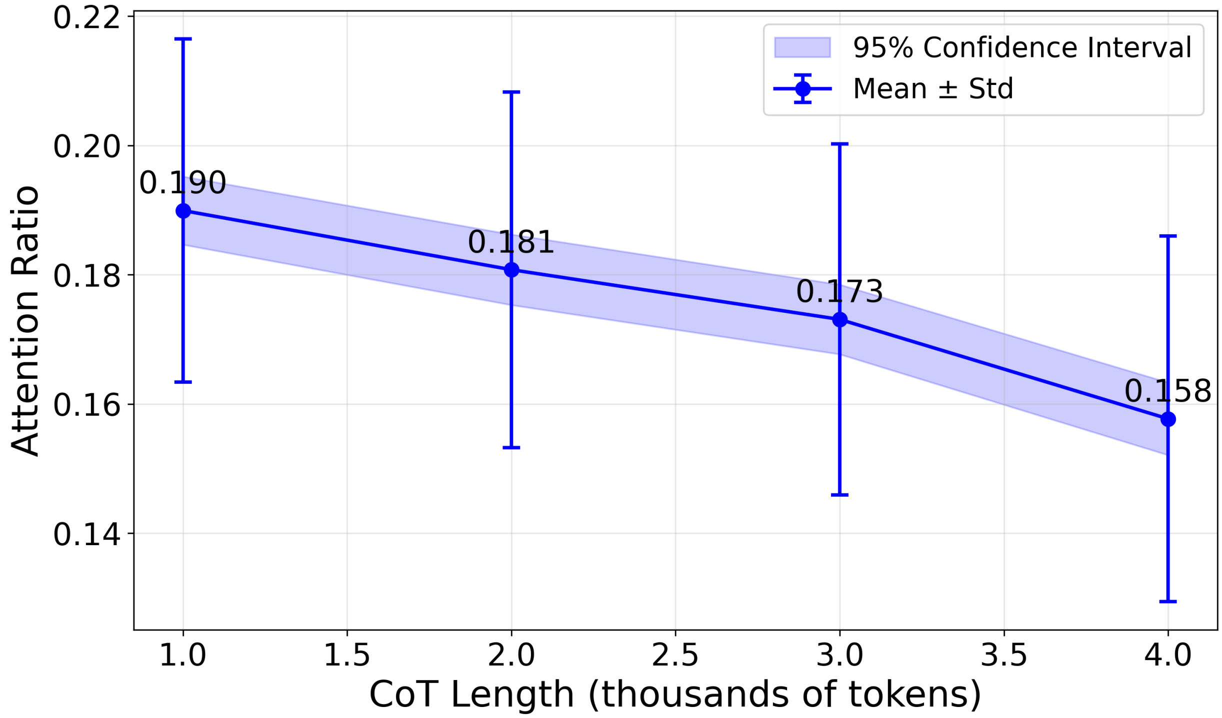

Figure 17. **Attention ratio trend across CoT lengths.** Ratio declines from 0.190 (1k) to 0.158 (4k). Longer CoT reduces attention paid to harmful instructions.

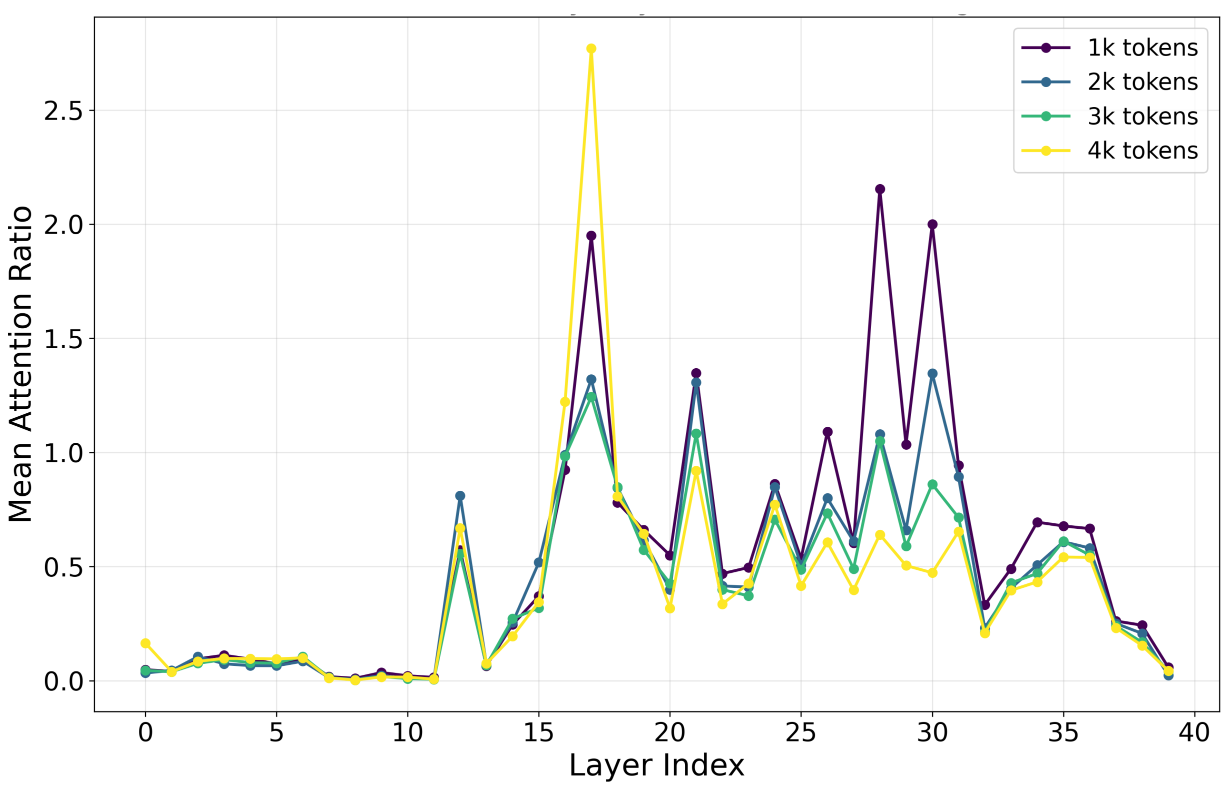

Figure 18. **Layer-wise attention ratio patterns (1k–4k CoT lengths).** The effect of CoT length is concentrated in layers 25–32, with layers 28 and 30 showing strongest differences.